

Q1  $\Rightarrow$  (i)  $(D^2 + 2D + 2)y = 4e^{-u} \sin^3(u)$

$$D^2 + 2D + 2 = 0$$

~~the~~ The characteristic equation

$$\gamma^2 + 2\gamma + 2 = 0$$

$$\gamma = \frac{-2 \pm \sqrt{4 - 4(1)(2)}}{2(1)} = -1 \pm i$$

Thus, the complementary sol<sup>n</sup> is

$$y_h = e^{-u} (C_1 \cos u + C_2 \sin u)$$

Solve the Particular sol<sup>n</sup>

~~Wronskian~~ Wronskian of  $e^{-u} \cos u$  &  $e^{-u} \sin u$  is

$$W = \begin{vmatrix} e^{-u} \cos u & e^{-u} \sin u \\ -e^{-u} \cos u - e^{-u} \sin u & -e^{-u} \sin u + e^{-u} \cos u \end{vmatrix}$$

$$= e^{-2u}$$

~~Find~~  
Compute the particular solution:

$$y'' = y_1 \int \frac{y_1 g(u)}{w} du - y_2 \int \frac{y_2 g(u)}{w} du$$

$$y_1 = e^{-u} \cos u, \quad y_2 = e^{-u} \sin u$$

$$g(u) = 4e^{-u} \sec^3(u)$$

$$y_p = e^{-u} \cos u \int \frac{e^{-u} \sin u (4e^{-u} \sec^3(u))}{e^{-2u}} du$$

$$= e^{-u} \sin u \int \frac{e^{-u} \cos u (4e^{-u} \sec^3(u))}{e^{-2u}} du$$

$$y_p = 4e^{-u} \cos u \int \sin u \sec^3(u) du$$

$$= 4e^{-u} \sin u \int \cos u \sec^3(u) du$$

$\rightarrow 4u + C$

$$= 4e^{-u} \cos u \int \tan(u) \sec^2(u) du$$

$$= 4e^{-u} \sin u \int \sec^2(u) du$$

$\rightarrow 4 \tan u + C$

$$y = y_h + y_p$$

$$y = e^{-u} (C_1 \cos u + C_2 \sin u) + 4u e^{-u} \cos u$$

$$= 4 \tan u \cdot e^{-u} \sin u$$

Ans



$$(ii) \quad (u^2 D^2 + u D - 1)y = \frac{1}{u^2}$$

This is a Cauchy-Euler eq<sup>n</sup>.

Let  $y_h = u^r$  be the sol<sup>n</sup> of homogeneous part

$$Dy = r u^{r-1}, \quad D^2 y = r(r-1) u^{r-2}$$

~~then~~ Substitute into the homogeneous eq<sup>n</sup>

$$u^2 (r(r-1) u^{r-2}) + u (r u^{r-1}) - u^r = 0$$

Simplifying:  $r(r-1) + r - 1 = 0$

$$r^2 - 1 = 0$$

$$\boxed{r = \pm 1}$$

Then the Complementary sol<sup>n</sup> is

$$\boxed{y_h = C_1 u + C_2 u^{-1}}$$

~~the~~  $y_1 = u \quad y_2 = u^{-1}$

$$W = \begin{vmatrix} u & u^{-1} \\ 1 & -u^{-2} \end{vmatrix} = -2u^{-1}$$

$-u^{-1} - u^{-1} = -2u^{-1}$

$$y_p = u \int \frac{1}{u} \cdot \frac{1}{u^2} (-2u^{-1}) du + \frac{1}{u} \int u \cdot \frac{1}{u^2} (-2u^{-1}) du$$

$$= -\frac{2u}{-3} \cdot \frac{1}{u^3} + \frac{1}{u} \left[ \frac{-2}{-1} \cdot \frac{1}{u} \right]$$

$$= \frac{2}{3u^2} + \frac{2}{u^2} = -\frac{4}{3u^2}$$

then  $y = c_1 u + c_2 u^{-1} - \frac{9}{3u^2}$  Ans

Similarly solve the all other question.

(ii)  $y'' + 8y' + 16y = 64 \cosh(4u)$   
 $D^2 + 8D + 16 = 64 \cosh(4u)$

Ⓢ Solve it similarly for other part also

Ⓢ Solve  $y' = y^2$ ,  $y(0) = 1$

$y = y_0 + \int_0^u f(t, y) dt$  or Picard  
iteration  
method

$y_0(u) = 1$

$y_1(u) = 1 + \int_0^u (t^2) dt = 1 + \frac{t^3}{3}$

$y_2(u) = 1 + \int_0^u (1+t)^2 dt = 1 + \left[ t + t^2 + \frac{t^3}{3} \right]_0^u$

$= 1 + u + \frac{u^2}{2} + \frac{u^3}{3}$

$y_3(u) = 1 + \int_0^u \left( 1 + t + t^2 + \frac{t^3}{3} \right) dt$

upto three iteration

$y \approx 1 + u + \frac{u^2}{2} + \frac{u^3}{3}$



$$(ii) \quad y' = y - y^2, \quad y(0) = 1$$

$$y(u) = y(0) + \int_0^u (y - y^2) dt$$

$$y_0(u) = 1$$

$$y_1(u) = 1 + \int_0^u (1 - 1^2) dt = 1 + \int_0^u (1 - 1) dt = 1$$

$$y_2(u) = 1 + \int_0^u (1 - 1^2) dt = 1$$

$$y_n(u) = 1$$

Suggesting that the sol<sup>n</sup> is constant  
 $y = 1$

$$(iii) \quad y' = uy + 2u - u^3, \quad y(0) = 0$$

$$y_n(u) = y(0) + \int_0^u f(t, y_n(t)) dt$$

$$f(u, y) = uy + 2u - u^3$$

$$y_0(u) = 0$$

$$y_1(u) = 0 + \int_0^u (t \cdot 0 + 2t - t^3) dt$$

$$y_1(u) = \left[ t^2 - \frac{t^4}{4} \right]_0^u = u^2 - \frac{u^4}{4}$$

$$y_2(u) = 0 + \int_0^u t \left( t^2 - \frac{t^4}{4} \right) + 2t - t^3 dt$$

$$= \left[ -\frac{t^5}{24} + t^2 \right]_0^u = -\frac{u^5}{24} + u^2$$

$$y(u) = \frac{u^2 - \frac{u^6}{29}}{29} \quad \underline{\underline{Ans}}$$